Adversarial self-audit of the manuscript, 2026-10-20

Draft of 2026-10-19

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The brief for this round was to attack our own product: find what is wrong, fix it, and re-verify. This note records the defects found, not the things that survived.

Method: the built PDF was read page by page against the artifacts; every quoted number was re-derived from its table; each caption was checked against the figure it describes; the class file was compared byte for byte with the official AGU repository; and the bibliography was checked field by field against the publisher deposits (separate report: `wrr_reference_audit_20261020.md`).

Defects found and fixed

id	severity	where	defect	fix
D1	Major, factual	§1, §2.3	“gap length and the share of the target window hidden correlate at −0.86”. The −0.86 is the correlation between gap length and the **warm-season fraction observed**; the correlation with hidden support is **+0.45** (development) and +0.47 (external).	both correlations now stated with the right quantities

id	severity	where	defect	fix
D2	Major, factual	§4.2	“the ordering of the five methods at 90 days is the same as at 30”. It is not: interpolation costs 0.34 °C at 30 days and 1.31 °C at 90, moving from second cheapest to most expensive.	the instability is now reported, and it strengthens the conditional claim
D3	Major, factual	§4.2	air regression was grouped with methods whose length effect is “not distinguishable from zero”; its interval is 0.13 °C [0.03, 0.22], which excludes zero.	climatology and smoothing only; air regression reported separately as small but distinguishable
D4	Major, factual	§4.4	“the interval includes zero at two of three seeds” for fitted minus bound on development. All three intervals include zero.	corrected to all three
D5	Major, overclaim	abstract, §4.4, conclusions	“no better than a calibrated certificate bound”. On the external panel seed 101 the fitted score is +1.6 points [0.2, 3.1], i.e. better, and the other two seeds cross zero; on development it is worse.	“not reliably better”, with both panels stated
D6	Major, overclaim	key point 3, abstract, §4.4, §5.3	“real calendars need no screen” and “already under the cap”. The point estimates are 2.6–4.1 %, but the intervals are wide and no cap is certified for a new archive.	“may need no screen”; “under the cap at every point estimate, although the intervals are wide”

id	severity	where	defect	fix
D7	Minor, scope	§1	the 1.31 °C error was attributed to “the complete station-years”, but it is measured on the designed outages placed on them.	attributed to the designed outages
D8	Major, missing	§3.1	the admitted object is a per-case policy over five reconstructors, but the paper never said what the policy chooses or where it fails.	added: donor 69 %, smoothed climatology 24 %, interpolation 6 %; the fallback fails at 38 % against 8 % for donor regression
D9	Minor, metadata	Figure captions	source-table paths were stale (the reconstructor caption pointed at `fig2_source.csv` after the figures were renumbered).	corrected to fig1/3/4/5 and the directory named once in the Open Research statement
D10	Minor, consistency	Figure 3 caption	“only linear interpolation retains a length effect” contradicted the air-regression interval.	caption now distinguishes the large interpolation effect from the small air effect
D11	Minor, stale	§5.2	the refused tail was still quoted as “35 to 55 % against 1 to 22 %” from an earlier binning.	40–55 % against 0.7–15 %
D12	Minor, vague	§5.3	“the retention differences are 7 to 30 points” mixed two contrasts.	fitted minus support 7–8 points; support minus length about 27
D13	Minor, contradiction	§5.2	“on the external panel the difference is small and interval-crossing” contradicted the +1.6 point interval quoted two paragraphs earlier.	“at most 1.6 points and crosses zero at two of the three seeds”

id	severity	where	defect	fix
D14	Minor, cross-reference	SI	“Figure 4c” after the figure renumbering; contents said “Tables S1 to S7” when there are eight; page count stale.	Figure 5c; Tables S1 to S8; counts updated
D15	Presentation	`paper/wrr/figures/`	nineteen legacy figure PDFs from earlier rounds were still in the include directory, triggering QA warnings and confusing the submission package.	only the six included files remain
D16	Presentation, brief	Figures 2 and S1	architecture/system-design schematics were matplotlib drawings, against the brief that they must be draw.io.	rebuilt as draw.io sources (`figures/wrr/drawio/`) and exported with the official draw.io desktop exporter (32.5.5, headless); the system design is now a main-text figure
D17	Integrity	bibliography	eight entries were duplicated across two `.bib` files, and seven entries had wrong fields (see the reference report).	one source of truth; fields corrected and re-verified

What the audit confirmed as correct

- Every retention, risk and hot-year number quoted in the text re-derived from its CSV: 39.6 / 66.7 / 72.9 / 74.0 % external; 30.5 / 60.4 / 72.7 / 70.4 % development; 2.6–4.1 % unscreened on real calendars; refusal 24.5–59.5 %.
- Every contrast quoted with an interval (fitted minus peak 8.0/7.3/7.0 and 10.3/10.1/9.7; fitted minus bound +1.6 and −2.6/−1.7/−2.5; subset contrasts).
- The effect sizes behind the mechanism: misalignment 0.28–0.70 °C, persistence 0.16–0.36 °C, within-support separation 0.7–15 % vs 40–55 %.
- The AGU class and its three supporting files are byte-identical to the official repository at the recorded commit (`PROVENANCE.md`).
- Figure fonts are embedded, no text outside the page, no line below 4.5 pt; the remaining QA warnings are box-model artifacts (superscripts, padding) verified against the renders.

What this audit does not establish

- We did not re-run every analysis from raw daily observations; the numbers are re-derived from the versioned case tables, not from the providers' servers.
- The external panel is not a sealed confirmation; the cross-fit statement is checked, not the independence of the original archive.
- The author block is still a placeholder, and the manuscript is correctly reported as not submission-ready.